Extraction of Fetal ECG by Logarithmic Hyperbolic Secant Adaptive Algorithm in Alpha-Stable Noise

Mengjia Wang, Deqiu Zhai, Jiacheng Zhang, Bo Ni, and Tao Liu

***Abstract*— Direct fetal electrocardiogram (FECG) plays a crucial role in assessing fetal health and monitoring pregnancy conditions. Extracting high-quality FECG signals from maternal abdominal electrocardiogram (AECG) recordings remains a significant challenge due to the low amplitude of the FECG, its overlap with the maternal electrocardiogram (MECG), and the potential exposure to impulsive noise in the real world. Adaptive filtering (AF) is an essential method for FECG extraction, however, its performance tends to degrade in the presence of impulsive noise, such as instrument interference. To address this limitation, we propose a novel AF algorithm based on a nonlinear logarithmic hyperbolic secant (LHS) cost function. Alpha-stable distribution is adopted to model the realistic noises due to its high scalability. To further enhance extraction accuracy and optimize the preset parameters, we introduce a hyperbolic tangent-like transformation and develop the improved logarithmic hyperbolic secant adaptive filtering (ILHSAF) algorithm. The proposed approach leverages the approximate linear interval of the LHS function to maximize the preservation of original FECG information within the AECG. We use the synthetic dataset FECGSYN as well as two real datasets, Daisy and NI-FECG, to evaluate the performance and our methods outperform other existing AF algorithms. The ILHSAF algorithm exhibits commendable performance in R-peak detection and full-wave analysis on both real-world datasets, indicating its effective denoising capability and robustness in FECG extraction. This advancement establishes a foundation for long-term maternal and fetal monitoring using portable devices, as the proposed algorithms are capable of real-time operation.**

***Index Terms*— FECG extraction, alpha-stable distribution, impulsive noise, logarithmic hyperbolic secant function, adaptive filtering.**

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I. INTRODUCTION

FETAL heart monitoring is crucial for establishing the health of both the mother and the baby during pregnancy. It not only enables the diagnosis of fetal cardiac function but also aids in evaluating the central nervous system function of the fetus, thereby improving the diagnostic accuracy of fetal distress and reducing perinatal mortality rates [1].

Current primary methods of fetal heart monitoring include Doppler ultrasound monitoring, magnetocardiography (MCG), and fetal electrocardiography (FECG). Doppler ultrasound estimates fetal heart rate by emitting high-frequency sound waves at 1–3 MHz, which requires sophisticated equipment and trained professionals for operation, making it unsuitable for long-term monitoring [2]. MCG adopts the magnetic resonance principle to detect morphological and physical movement phenomena of the heart, which is more refined than Doppler ultrasound [3]. However, magnetic resonance devices require specialised magnetically shielded environments and cannot be used for prolonged monitoring or mobile device development [4]. FECG provides a direct reflection of fetal health with high diagnostic value and is a sensitive indicator for the early diagnosis of intrauterine fetal distress. Morphological and temporal features (detected by QRS complex) of FECG provide important information about fetal health. The detection success rate for fetal intrauterine abnormalities using FECG exceeds 90% [5]. The measurement of FECGs can be categorized into two methods: invasive and non-invasive. Invasive FECG involves placing electrodes on the fetal scalp during labor, which poses risks to the fetus and is unsuitable for long-term monitoring or perinatal surveillance. Non-invasive FECG (NIFECG) use maternal abdominal ECG signal for FECG extraction and allow real-time monitoring of fetal cardiac activity [6]. NIFECG has recently shown promise for fetal heart monitoring but faces two major challenges:

- FECG is smaller in amplitude compared to the maternal electrocardiogram (MECG) and usually overlaps MECG, making it difficult to distinguish between them [7].
- The abdominal electrocardiogram (AECG) obtained from the abdomen of a pregnant often competes with contamination from various noise sources, including myoelectric signal interference [8] and instrumental interference [9]. These noises are typically non-Gaussian in nature and accompanied by strong impulses [10].

Therefore, how to obtain a clear and accurate FECG recording has become a difficult problem in NIFECG extraction. Presently, the common algorithms suggested by researchers for NIFECG extraction include wavelet transform [11], blind source separation [12], [13], deep learning [9], [14], adaptive filtering and others [8], [15], [16].

Adaptive filtering (AF) signifies an important mean of NIFECG extraction with lower computational complexity and easier implementation [17]. The filter is trained on variable coefficients using an objective function to remove the maternal ECG component from AECG signal [18]. Traditional AF algorithms, like the least mean square (LMS) algorithm, mostly rely on the mean square error (MSE) criterion as the objective function. A normalized least mean square (NLMS) algorithm was used in [19] for FECG extraction and fetal heart rate processing. In [20], the authors implemented an improved proportionate normalized least mean square (IPNLMS) algorithm for FECG extraction, which updates each filter coefficient independently. Both algorithms effectively detected the R-peaks of the FECG extracted, achieving detection accuracy of 92.86% [19] and 93.75% [20], respectively. However, they still obey the MSE criterion, the second order statistic MSE has a poor fit in non-Gaussian environments, which often leads to degradation of algorithmic performance and inappropriate approximation to the real system under impulsive noises [15].

To address this issue, several nonlinear hyperbolic functions have been proposed as the cost function for AF algorithms. [21] proposed a novel lncosh function. The distinctive feature of this function lies in its adjustable parameter λ, which allows the lncosh cost function to approximate a hybrid of the MSE and mean absolute error (MAE) criteria. Specifically, when λ approaches 0, the cost function approximates MSE, when λ tends toward ∞, it approximates MAE. AF algorithms based on lncosh benefits from the saturation effect of the hyperbolic tangent function, which suppresses outliers and exhibits strong robustness against impulsive noise. With further researches, derivative algorithms of lncosh have been successfully applied in various fields, including system identification [22], DOA channel estimation [23], and nonlinear adaptive noise cancellation [24]. These algorithms are characterized by strong robustness and high resistance to interference. Recently, hyperbolic secant functions have attracted the scholars' attention [24]–[29]. AF algorithms based on the hyperbolic secant function further improves filtering performance, including lower steady-state error, enhanced robustness, and better convergence [30]. A generalized hyperbolic secant adaptive filter (GHSAF) based on the nearest kronecker product decomposition (NKP) was proposed in [31]. The proposed algorithm shown the improved convergence performance for robust system identification under non-Gaussian noise environments. To improve the performance of the kernel adaptive function in impulsive noises, [32] combined the hyperbolic secant function with the minimum kernel risk-sensitive loss algorithm and proved its effectiveness through experiments.

Inspired by the nonlinear hyperbolic cost function, we propose an AF algorithm based on a novel logarithmic hyperbolic secant (LHS) function to overcome the noise adulteration problem in NIFECG extraction. Meanwhile, we introduce a hyperbolic tangent-like transformation [33] to ensure the integrity of the FECG and the improved logarithmic hyperbolic secant adaptive filtering (ILHSAF) algorithm is derived. The α-stable distribution with heavy-tailed probability density function is used to verify the robustness of the algorithm, since it is commonly recommended for modeling impulsive noise [34]. We evaluate the efficiency of our methods with one synthetic dataset and two real-world datasets. The proposed algorithms show significant improvements in NIFECG extraction when compared to several existing AF methods.

The main contributions of this work are summarized as follows:

1) We propose a novel logarithmic hyperbolic secant cost function and derive the corresponding algorithm to address the issue of impulsive noise contamination in NIFECG extraction.
2) A hyperbolic tangent-like transformation is introduced to tackle the challenges of the FECG's low amplitude and overlap with MECG, ensuring the integrity of FECG information.
3) The proposed method is evaluated from three aspects. We assess the quality of the extracted FECG in terms of graphical responses, R-peak detection results and full-wave analysis, respectively.

The structure of this paper is organized as follows. In Section II, the signal model, noise model and previous related studies are introduced and reviewed. Section III details the proposed LHS cost function and the ILHSAF algorithm. Furthermore, the convergence and computational complexity of the proposed methods are analyzed. In Section IV, experimental results on real and synthetic datasets are discussed. Section V concludes the main contribution of our work.

List of Abbreviations

FECG	Fetal electrocardiography
AECG	Abdominal electrocardiogram
MECG	Maternal electrocardiogram
NIFECG	Non-invasive FECG
AF	Adaptive filtering
LMS	Least mean square
NLMS	Normalized least mean square
IPNLMS	Improved proportionate normalized least mean square
LHS	Logarithmic hyperbolic secant
ILHS	Improved logarithmic hyperbolic secant
LHSAF	Logarithmic hyperbolic secant adaptive filtering
ILHSAF	Improved logarithmic hyperbolic secant adaptive filtering
SGD	Stochastic gradient descent
Llncosh	Least lncosh
ALI	Approximately linear interval
SI	Saturation interval
Se	Sensitivity
PPV	Positive predictive value

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II. Related Works

A. Signal model

The AECG signal recorded from a pregnant individual comprises multiple components: the FECG signal, the maternal electrocardiogram component denoted as $m(n)$ and the additive noise $v(n)$. Research presented in [35] indicates that the amplitude of $m(n)$ exhibits a correlation with MECG, typically accounting for approximately 20% of the amplitude of MECG. Consequently, the expression of AECG is given by:

$$\begin{aligned} \text{AECG} &= \text{FECG} + m(n) + v(n) \\ &= \text{FECG} + 0.2\text{MECG} + \text{noise}. \end{aligned} \tag{1}$$

Fig. 1 shows the signal model of FECG extraction with adaptive filtering [36]. The input signal $\boldsymbol{x}(n)$ corresponds to the thorax signal MECG and the abdominal signal AECG is the desired signal $d(n)$. The output signal $y(n)$ is derived from $\boldsymbol{x}(n)$ in conjunction with the filter coefficients $\boldsymbol{w}(n)$, denoted as:

$$y(n) = \boldsymbol{w}^{\mathsf{T}}(n)\boldsymbol{x}(n), \tag{2}$$

where $\boldsymbol{x}(n) = [x(n), x(n-1), \cdots, x(n-L+1)]^{\mathrm{T}}$, $\boldsymbol{w}(n) = [w_1(n), w_2(n), \cdots, w_L(n)]^{\mathrm{T}}$ and L is the filter order. The filter modifies the weight coefficients $\boldsymbol{w}(n)$ in an adaptive manner based on the error $e(n)$ between $y(n)$ and $d(n)$. This adjustment aims to minimizes the objective function $\boldsymbol{J}(e(n))$, ultimately yielding the extracted FECG signal. The theoretical representation of this process is provided below:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) - \mu \nabla \boldsymbol{J}(e(n)), \tag{3}$$

$$e(n) = d(n) - y(n), \tag{4}$$

$\mu > 0$ is the adjustment step of the filter.

B. Noise model

In modern applications, signals often exhibit sharp spikes are referred to as non-Gaussian noise or impulsive noise. They are characterized by a distribution that diminishes at a slower rate compared to Gaussian noise. The α-stable distribution has proven to be effective for modeling such noise signals. This distribution originates from a generalized central limit theorem (GCLT) and serves as an extension of the Gaussian distribution. The probability density function (PDF) of the α-stable distribution lacks a closed-form expression [37], and $S_\alpha(\beta, \gamma, \delta)$ is typically represented by its characteristic function:

$$\phi(t) = \exp\{\mathrm{j}\delta t - \gamma^\alpha |t|^\alpha [1 + \mathrm{j}\beta \operatorname{sgn}(t) \varepsilon(t, \alpha)]\}, \tag{5}$$

where

$$\varepsilon(t, \alpha) = \begin{cases} \tan(\pi\alpha/2), & \alpha \neq 1 \quad \text{(6a)} \\ (-2/\pi)\ln|t|, & \alpha = 1. \quad \text{(6b)} \end{cases}$$

$\alpha \in (0, 2]$ represents the characteristic exponent that controls the thickness of the PDF tail. A low value of α signifies a high degree of impulsiveness, which results in a heavier PDF tail [38]. $\beta \in [-1, 1]$ represents the symmetry parameter and describes the skewness of the distribution. $\gamma \in (0, +\infty)$ is the dispersion parameter, analogous to the variance in the Gaussian distribution. $\delta \in (-\infty, +\infty)$ is the location parameter, specifically, for $\alpha \in (0, 1)$ or $\alpha \in [1, 2]$, δ corresponds to the median or mean, respectively. When $\beta = 0$, the α-stable distribution is symmetric around δ and is designated as the symmetric α-stable (SαS) distribution [39]. When α = 2, the SαS distribution is equivalent to the Gaussian distribution.

C. Related algorithms

1) MSE-Based Algorithms: The LMS algorithm is a prominent representative of MSE-based algorithms and has gained widespread popularity as an AF algorithm due to its versatility and simplicity. Its key innovation is the utilization of the instantaneous squared error to approximate the MSE. Grounded in the stochastic gradient descent (SGD) criterion, the weight update formula for the LMS algorithm is derived as follows:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu_{\text{LMS}} e(n)\boldsymbol{x}(n). \tag{7}$$

(7) is easy to implement and does not require complex computational processes. However, LMS has certain limitations, particularly when $\boldsymbol{x}(n)$ possesses a high amplitude. In such cases, the contribution of $\boldsymbol{x}(n)$ in the term $\mu e(n)\boldsymbol{x}(n)$ becomes disproportionately significant, leading to an amplification of gradient noise, as $\boldsymbol{x}(n)$ is perceived as a noisy signal. To overcome this challenge, NLMS algorithm was developed [40]. NLMS algorithm serves as a practical application of the principle of minimizing interference. (8) illustrates the recursive model of the NLMS algorithm to adjust the filter weights, with the parameter o_{NLMS} to prevent division by zero.

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \frac{\mu_{\text{NLMS}} e(n)\boldsymbol{x}(n)}{\boldsymbol{x}(n)^{\mathsf{T}}\boldsymbol{x}(n) + o_{\text{NLMS}}}, \tag{8}$$

when $\mu_{\text{NLMS}} \in (0, 2)$, the NLMS algorithm converges in the mean-square sense. A drawback of the NLMS algorithm is that its convergence rate slows down when the input signals exhibit high correlation, and its complexity is higher than the LMS algorithm [41]. Subsequent research led to the development of the IPNLMS algorithm [42], which introduces the concept of assigning distinct adaptive step sizes to each filter coefficient. This approach reallocates the adaptive gain across all the coefficients, placing greater emphasis on larger coefficients to enhance their convergence rate. Consequently, active coefficients are adjusted more rapidly compared to inactive ones. The IPNLMS algorithm is described by:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \frac{\mu_{\text{IPNLMS}} e(n)\boldsymbol{P}(n)\boldsymbol{x}(n)}{\boldsymbol{x}(n)^{\mathsf{T}}\boldsymbol{P}(n)\boldsymbol{x}(n) + o_{\text{IPNLMS}}}, \tag{9}$$

where, $\boldsymbol{P}(n) = \text{diag}[p_1(n), p_2(n), \cdots, p_L(n)]^{\mathsf{T}}$ is a diagonal matrix of the step sizes for each tap, o_{IPNLMS} is the regularisation parameter. The diagonal elements of $\boldsymbol{P}(n)$ are calculated as follows:

$$p_l(n) = \frac{1 - \alpha_{\text{IPNLMS}}}{2L} + \frac{(1 + \alpha_{\text{IPNLMS}})|w_l(n)|}{2L\|\boldsymbol{w}(n)\| + \lambda_0}. \tag{10}$$

$\alpha_{\text{IPNLMS}} \in [-1, 1]$ is regarded as the proportional term that integrates the first and second terms of (10). λ_0 represents a tiny positive constant. In our simulations, IPNLMS always performs better than NLMS when α_{IPNLMS} = 0.2 and o_{IPNLMS}= λ_0 = 10^{-10}. However, IPNLMS needs to perform more computations, increasing its computational complexity.

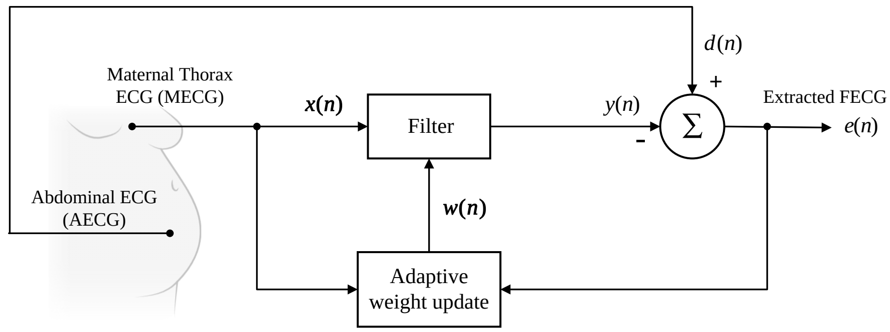

Fig. 1. Block diagram of FECG extraction with adaptive filters.

2) The Least Lncosh Algorithm: The MSE-based algorithms discussed above are based on the second-order statistics, which has a poor fit in real system under non-Gaussian noise environments [22]. In contrast to the MSE criterion, [21] states that the lncosh function, which is characterised by the natural logarithm of hyperbolic cosine, does not rely on particular noise distribution assumptions. This function demonstrates good performance in the presence of impulsive noises. The lncosh cost function is defined as:

$$\boldsymbol{J}_{\text{Lncosh}}(e(n)) = \mathbf{E}\left[\frac{1}{\lambda}\ln(\cosh(\lambda e(n)))\right]. \tag{11}$$

According to the SGD, the weight updating equation of the robust least Lncosh (Llncosh) algorithm is derived as:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu_{\text{Llncosh}}\tanh(\lambda e(n))\boldsymbol{x}(n), \tag{12}$$

$\mu_{\text{Llncosh}} > 0$. Researches have indicated that the amplitude of the received signal affected by impulsive noise can be effectively constrained through the application of nonlinear preprocessing functions [37], [38]. Among these, the hyperbolic tangent function is the most frequently used [43]. (12) illustrates that the Llncosh algorithm demonstrates superior performance compared to the MSE-based algorithms in the context of impulsive noise suppression. The hyperbolic tangent function, denoted as $\tanh(\cdot)$, exhibits the property that as the $\lambda e(n)$ increases, $\tanh(\lambda e(n))$ approaches $\text{sgn}(\lambda e(n))$, effectively bounding large errors $e(n)$ [28]. This behavior functions similarly to an amplitude limiter.

III. Approaches

A. Logarithmic Hyperbolic Secant Cost Function

Drawing inspiration from the nonlinear preprocessing amplitude-limiting method [44], this study proposes an innovative LHS function to serve as the cost function and develops a logarithmic hyperbolic secant adaptive filtering (LHSAF) algorithm. To enhance its applicability for the specific context of NIFECG extraction, we introduce a hyperbolic tangent-like transformation with predetermined parameters into the proposed algorithm. This procedure aims to maximize the preservation of information from the original FECG signal while simultaneously mitigating the impact of impulsive noise. The proposed LHS cost function is given by:

$$\boldsymbol{J}(e(n)) = \mathbf{E}\left[-\frac{1}{\lambda}\ln(1+\text{sech}(\lambda e(n)))\right], \tag{13}$$

where, $\ln(\cdot)$ is the natural logarithm, $\text{sech}(\cdot)$ denotes the hyperbolic secant function, $\lambda > 0$. The first derivative of this function can be expressed as:

$$\begin{aligned}\boldsymbol{J}'(e(n)) &= \frac{\partial \boldsymbol{J}(e(n))}{\partial e(n)} \\ &= \frac{\text{sech}(\lambda e(n))\tanh(\lambda e(n))}{1+\text{sech}(\lambda e(n))}.\end{aligned} \tag{14}$$

The relationship between $e(n)$ and the gradient of the conventional MSE-based algorithms is proportional, owing to its use of the second-order mean square error criterion [24]. A sudden significant change in $e(n)$ will results in a corresponding adjustment in $\boldsymbol{w}(n)$, which can cause the algorithm to diverge.

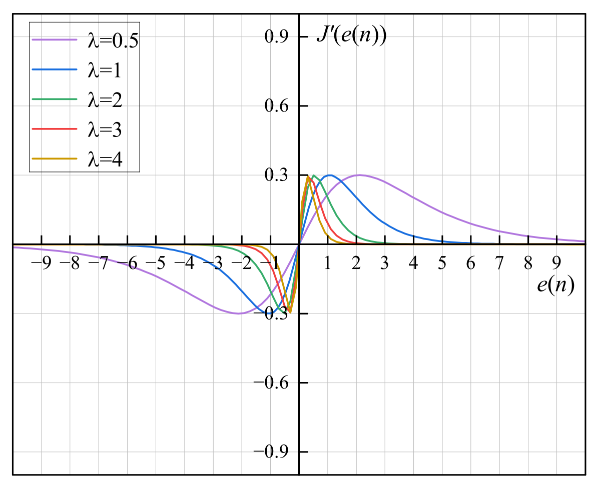

Fig. 2. The gradient of the novel cost function vs. $\boldsymbol{e(n)}$ for different values of $\boldsymbol{\lambda}$.

Fig. 2 presents the gradient of $\boldsymbol{J}(e(n))$ with respect to $e(n)$ for different values of λ. It illustrates that the proposed function exhibits odd symmetry and features a distinct peak. When $e(n)$ is large, the amplitude of the gradient is infinitely close to 0, indicating its robustness against outliers. Moreover, it is noteworthy that the peak value of the gradient consistently remains below 0.3, regardless of the value of λ. This suggests that although impulsive noise outliers may lead to considerable errors, the amplitude-limiting preprocessing effect of the cost function effectively mitigates their impact. This characteristic aligns with the expectation of a robust algorithm.

For tractability, the cost function can be approximated using

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the instantaneous form of (13) [21], which is expressed as:

$$\bar{\boldsymbol{J}}(e(n)) = -\frac{1}{\lambda}\ln(1+\operatorname{sech}(\lambda e(n))), \tag{15}$$

the derivative to the $\boldsymbol{w}(n)$ is derived as:

$$\begin{aligned}&\frac{\partial \bar{\boldsymbol{J}}(e(n))}{\partial \boldsymbol{w}(n)} \\ &= \frac{1}{\lambda}\frac{\partial(-\ln(1+\operatorname{sech}(\lambda e(n))))}{\partial \boldsymbol{w}(n)} \\ &= \frac{1}{\lambda}\frac{\partial(-\ln(1+\operatorname{sech}(\lambda e(n))))}{\partial(1+\operatorname{sech}(\lambda e(n)))}\frac{\partial(1+\operatorname{sech}(\lambda e(n)))}{\partial \boldsymbol{w}(n)} \\ &= \frac{1}{\lambda}\frac{-1}{1+\operatorname{sech}(\lambda e(n))}\frac{\partial(1+\operatorname{sech}(\lambda e(n)))}{\partial \boldsymbol{w}(n)},\end{aligned} \tag{16}$$

where

$$\begin{aligned}&\frac{\partial(1+\operatorname{sech}(\lambda e(n)))}{\partial \boldsymbol{w}(n)} \\ &= \frac{\partial(1+\operatorname{sech}(\lambda e(n)))}{\partial \lambda e(n)}\frac{\partial \lambda e(n)}{\partial \boldsymbol{w}(n)} \\ &= \tanh(\lambda e(n))(-\operatorname{sech}(\lambda e(n)))(-\lambda \boldsymbol{x}(n)) \\ &= \lambda \tanh(\lambda e(n))\operatorname{sech}(\lambda e(n))\boldsymbol{x}(n),\end{aligned} \tag{17}$$

so

$$\frac{\partial \bar{\boldsymbol{J}}(e(n))}{\partial \boldsymbol{w}(n)} = -\frac{\tanh(\lambda e(n))\operatorname{sech}(\lambda e(n))}{1+\operatorname{sech}(\lambda e(n))}\boldsymbol{x}(n). \tag{18}$$

According to SGD method, the weight update rule derived from the LHS function is given as:

$$\begin{aligned}\boldsymbol{w}(n+1) &= \boldsymbol{w}(n) - \mu\frac{\partial \bar{\boldsymbol{J}}(e(n))}{\partial \boldsymbol{w}(n)} \\ &= \boldsymbol{w}(n) + \mu\frac{\tanh(\lambda e(n))\operatorname{sech}(\lambda e(n))}{1+\operatorname{sech}(\lambda e(n))}\boldsymbol{x}(n).\end{aligned} \tag{19}$$

(19) is henceforth called the LHSAF algorithm.

B. Improved LHS Adaptive Filtering Algorithm

The capture of clean FECG is difficult since it is a relatively weak signal, which constitutes less than 20% of MECG in amplitude and is always mixed with the rest components of AECG. Noise characterized by Gaussian properties can be suppressed using algorithms based on second-order statistics. In the presence of noise following an α-stable distribution, our approach prioritizes the preservation of the signal of interest (SOI) while permitting flexible modulation of both SOI and the noise. To this end, we propose the ILHSAF algorithm which incorporates a hyperbolic tangent-like transformation.

Fig. 3 displays the graph of $\tanh(\lambda x)$ for different value of λ. We can observe that the function exhibits an approximately linear behavior in the vicinity of $x = 0$. As $|x|$ increases, the amplitude of $\tanh(\lambda x)$ grows gradually and approaches a saturation level. Thus, the curve of the function can be categorized into two regions: the approximately linear interval (ALI) near the origin and the saturation interval (SI) away from the origin. The point x_0, where the curvature of the function reaches its maximum, delineates the boundary of the ALI. Variations in the value of λ result in either an extension or compression of the linear interval. At this point,

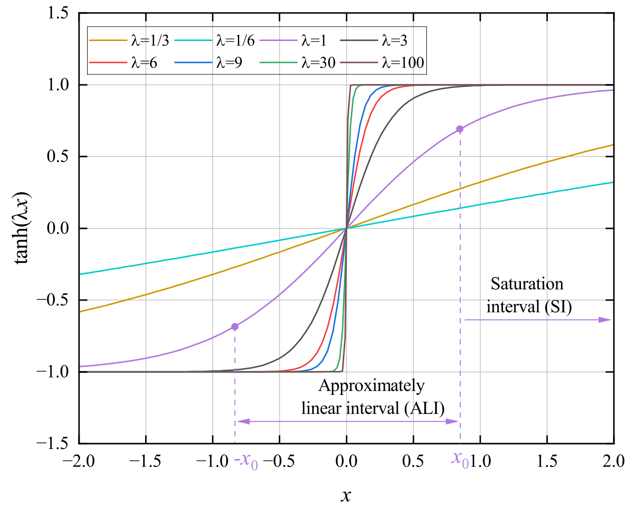


Fig. 3. The behavior of $\tanh(\lambda x)$ for different value of λ.

the corresponding hyperbolic tangent transformation is defined as:

$$\kappa(x) = \frac{\mathrm{H}}{x_0}\tanh(\frac{x_0 x}{\mathrm{H}}), \quad \lambda = \frac{x_0}{\mathrm{H}}. \tag{20}$$

H is the scale parameter and x_0/H is named the transformation factor. Relevant studies pointed that the selection of the scale parameter H in hyperbolic tangent-like methods is only associated with the SOI [43]. Based on the amplitude of SOI, (20) can be interpreted as a limiter that maintains the SOI within a preset threshold, simultaneously constraining any noise that surpasses the threshold. Notably, the threshold is determined only by the scale parameter H.

The comparison between Fig. 2 and Fig. 3 clearly reveals the similarities between the functions $\boldsymbol{J}'(e(n))$ and $\tanh(\lambda x)$, both exhibit approximately linear behavior near origin. However, the difference lies in that $\boldsymbol{J}'(e(n))$ exhibits a decaying characteristic as it deviates from origin, which is referred to as the attenuation interval (AI). Therefore, when calculating the ALI of $\boldsymbol{J}'(e(n))$, the point of maximum curvature may fall within the AI. It means that the ALI cannot be obtained solely by locating the point of maximum curvature. It is important to highlight that $\boldsymbol{J}'(e(n))$ is both monotonic and bounded within the ALI. Hence, the ALI can be determined by identifying the points at which the function attains its extrema within the interval.

Let the second derivative of (14) equal to 0, yielding

$$\begin{aligned}\frac{\partial^2 \bar{\boldsymbol{J}}(e(n))}{\partial e^2(n)} &= \frac{\lambda(\operatorname{sech}^4(\lambda e(n)) + \operatorname{sech}^3(\lambda e(n)))}{(\operatorname{sech}(\lambda e(n))+1)^2} \\ &\quad - \frac{\tanh^2(\lambda e(n))\operatorname{sech}(\lambda e(n))}{(\operatorname{sech}(\lambda e(n))+1)^2} \\ &= 0,\end{aligned} \tag{21}$$

in accordance with the hyperbolic tangent transformation, the function can be approximated by a linear function with slope k intersecting the origin within the ALI. A slope value closer to 1 indicates more effective preservation of the SOI information [43]. Numerical calculations reveal that when $\lambda = 3$, the slope approaches to 1 most closely. In this study, we take $\lambda = 3$ as

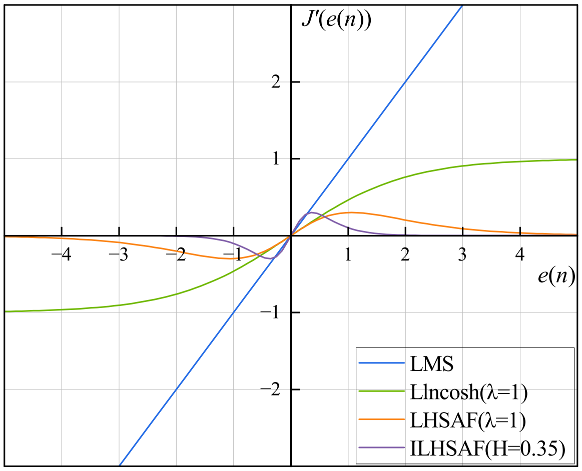


Fig. 4. Gradient of different AF algorithms versus $\boldsymbol{e(n)}$.

an example. Substituting into (21) yields:

$$e_0(n) = 0, \quad e_1(n) \approx -0.35, \quad e_2(n) \approx 0.35. \tag{22}$$

So, the ALI of the $\boldsymbol{J}^{'}(e(n))$ is approximately $[-0.35, 0.35]$, which varies with λ. The hyperbolic tangent-like transformation is derived as:

$$\psi(e(n)) = \frac{\tanh(\frac{0.35}{\mathrm{H}}e(n))\,\mathrm{sech}(\frac{0.35}{\mathrm{H}}e(n))}{1 + \mathrm{sech}(\frac{0.35}{\mathrm{H}}e(n))}. \tag{23}$$

In our signal model, the SOI is the FECG component of the AECG, whose amplitude range remains indeterminate. Prior researches indicated that the amplitude of MECG is several times greater than that of the FECG, with estimates suggesting it may be roughly 5 to 10 times [45], [46]. Therefore, it is reasonable to approximate the amplitude range of FECG based on that of the MECG. In this study, the estimated amplitude of the FECG is represented as $\hat{s}(n) = 0.1\mathrm{MECG}$, with the range of H defined as follows:

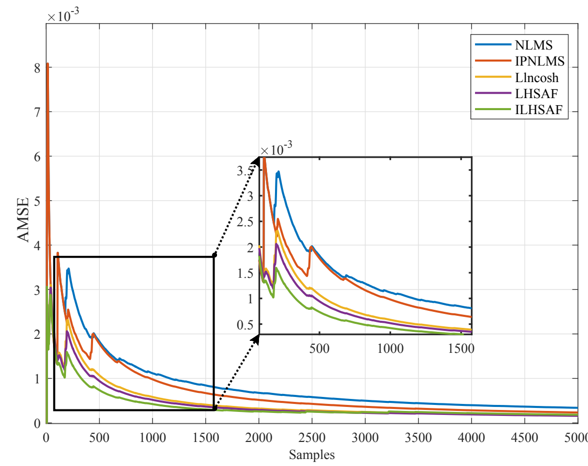
$$\mathrm{H} = [0, \max\{|\min(\hat{s}(n))|, |\max(\hat{s}(n))|\}], \tag{24}$$

by introducing the hyperbolic tangent-like transformation, the update equation of the ILHSAF algorithm is modified to:

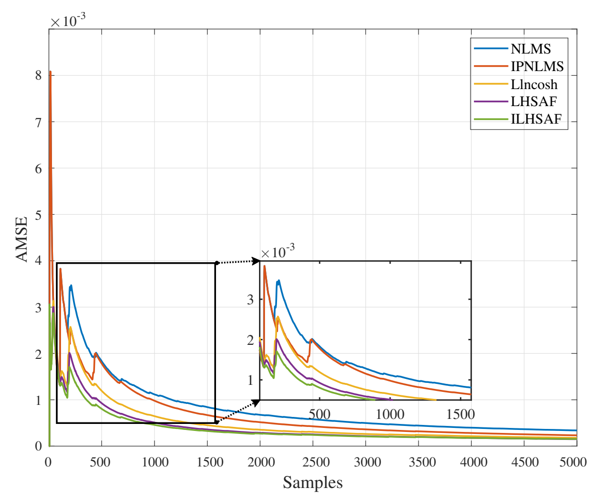
$$\begin{aligned}\boldsymbol{w}(n+1) &= \boldsymbol{w}(n) \\ &\quad + \mu\frac{\tanh(\frac{0.35}{\mathrm{H}}e(n))\,\mathrm{sech}(\frac{0.35}{\mathrm{H}}e(n))}{1 + \mathrm{sech}(\frac{0.35}{\mathrm{H}}e(n))}\boldsymbol{x}(n).\end{aligned} \tag{25}$$

To provide an intuitive understand of the robustness mechanism of the proposed algorithms, Fig. 4 shows the gradient of the cost function versus the error for several AF algorithms. For the conventional LMS algorithm, a clear proportional relationship between the error and its gradient is observed, attributed to the use of the MSE criterion. Consequently, abrupt changes in $e(n)$ can cause large weight updates, potentially leading to divergence of the algorithm. Llncosh algorithm exhibits a saturation effect in its gradient for large errors, which is not conducive to coping with sudden changes. Notably, the gradients of LHSAF and ILHSAF approach zero as the error increases, effectively suppressing the impact of impulsive noise and thereby enhancing robustness. Since the variation of error and weights should be minimized at steady state [46], the proposed algorithm can achieve more precise control effects. Moreover, we introduce a hyperbolic tangent-like transformation to address the challenges posed by the low amplitude of FECG and its overlap with MECG, thereby ensuring the integrity of FECG information.

C. Convergence and Complexity



(a) $\alpha = 2.0$



(b) $\alpha = 1.4$

Fig. 5. Convergence behaviour of algorithms in different noise backgrounds.

Fig. 5 shows the average iterative MSE (AMSE) of each algorithm over 100 independent trials for different values of α. It is observed that the ILHSAF algorithm demonstrates a consistently faster convergence rate compared to the rest several existing algorithms and also displays a comparatively lower AMSE. This suggests that the ILHSAF algorithm possesses superior robustness against impulsive noise. The parameters used in the simulations are detailed in Table I and are selected through Monte Carlo experiments to achieve a relatively optimal filtering performance.

Table II presents the mathematical computational complexity of the algorithms to execute a single iteration. It is evident that the computational complexity of the LHSAF and ILHSAF

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TABLE I
SIMULATION PARAMETERS

Algorithms	Simulation Parameters									
	$S_{2.0}(0,\gamma,0)$					$S_{1.4}(0,\gamma,0)$				
	L	μ	α_{IPNLMS}	λ	H	L	μ	α_{IPNLMS}	λ	H
NLMS	180	0.1	-	-	-	180	0.1	-	-	-
IPNLMS	200	0.04	0.2	-	-	200	0.04	0.2	-	-
Llncosh	7	0.01	-	10	-	7	0.01	-	10	-
LHSAF	7	0.02	-	8	-	7	0.02	-	10	-
ILHSAF	7	0.02	-	-	0.04	7	0.03	-	-	0.15

TABLE II
COMPUTATIONAL COMPLEXITY

Algorithms	$\times$	$+$	tanh	sech	$\div$
LMS	2L+1	2L	-	-	-
NLMS	3L+1	3L	-	-	1
IPNLMS	6L+4	4L+3	-	-	3
Llncosh	2L+2	2L	1	-	1
LHSAF	2L+3	2L+1	1	1	1
ILHSAF	2L+3	2L+1	1	1	2

algorithms is significantly lower than that of the NLMS and IPNLMS algorithms, while exhibiting a similar complexity level to that of the Llncosh algorithm.

IV. EXPERIMENTS AND RESULTS

A. Dataset

1) Synthetic Data: We use the synthetic dataset FECGSYN from PhysioNet [47] to evaluate the proposed algorithms. This dataset includes simulated signals from 10 pregnant subjects under 7 different physiological events and provides annotated R-peak locations of the FECG signals. Each set of recordings contain direct FECG signals as well as MECG signals. Five distinct signal-to-noise ratios (0, 3, 6, 9, and 12 dB) are employed to simulate real-world conditions, with a sampling frequency of 250 Hz. We select all traces in the dataset that were identified as case zero (no event occurred) with a noise level of 6 dB for evaluation.

2) Real Data: Additionally, we use two real-world datasets to validate the R-peak detection accuracy of the proposed algorithms. The first one is Daisy (Database for the Identity of Systems) [48], which consists of 8 electrode outputs recorded from a single pregnant participant, 3 for chest signals and 5 for abdominal signals. All data were 10 seconds long and sampled at 250 Hz.

The second one is the Non-Invasive Fetal ECG Database (NI-FECG) [49] from Physionet. This dataset contains 55 multichannel MECG signals collected from a single subject during the gestational period of 21 to 40 weeks. Each set of recordings contains 2 MECG signals and 3 to 4 mixed abdominal signals. The data were sampled at 1 kHz and band-pass filtered (0-100 Hz) during acquisition. In addition, NI-FECG provides R-peak reference annotation information but not FECG signals directly from the fetal scalp.

B. Evaluation statistics

For the synthetic dataset, the statistics we use for evaluation are: signal-to-noise ratio (SNR), root mean square error (RMSE) and percentage root mean square difference (PRD). They are used to evaluate the performance of algorithms in FECG extraction, with a known FECG signal provided as a reference. The expressions are as follows:

$$\mathrm{SNR} = 10\log\left(\frac{\sum_{n=1}^{L}|e(n)|^2}{\sum_{n=1}^{L}|s(n)-e(n)|^2}\right) \tag{26}$$

$$\mathrm{RMSE} = \sqrt{\sum_{n=1}^{L}\frac{(s(n)-e(n))^2}{L}} \tag{27}$$

$$\mathrm{PRD} = \sqrt{\frac{\sum_{n=1}^{L}(s(n)-e(n))^2}{\sum_{n=1}^{L}s^2(n)}} \times 100\% \tag{28}$$

where $s(n)$ is the direct FECG provided by the dataset and $e(n)$ is the extracted FECG.

For real datasets, the identification of R-peaks in ECG signals serves as a primary metric for assessing the efficacy of FECG extraction methods. To detect R-peaks, we applied the modified Pan-Tompkins algorithm [50]. [12] pointed that the reference markers are set to limit the minimum distance between R-peaks, commonly referred to as the RR interval. Physiologically, the RR interval is not expected to be less than 200 ms, but for the fetuses, the average heart rate varies from 120 to 165 beats per minute, corresponding to the RR intervals shorter than this threshold. Thus, we adjust the minimum distance between each R-wave reference marker used in Pan-Tompkins algorithm. The extracted FECG signals are subjected to the modified Pan-Tompkins algorithm to locate the fetal R-peak positions. The results are analysed by sensitivity (Se) [51], positive predictive value (PPV) [51] and F1-score [51] in comparison to the R-peak reference annotations provided. These metrics can be defined in terms of TP (True Positive), which means that the position of the fetal R-peak is correctly detected within ±10ms of the reference annotation, FP (False Positive) is the number of false positives, meaning the number of non-R-peak positions that are detected as R-peaks, and FN (False Negative), implying the number of missed detections. The presence of FP and FN indicates a

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faulty condition.

$$\mathrm{Se} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \times 100\% \tag{29}$$

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \times 100\% \tag{30}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{PPV} \times \mathrm{Se}}{\mathrm{PPV} + \mathrm{Se}} \tag{31}$$

C. Additive Noise Generation

The additive noise $v(n)$ is characterized by the α-stable distribution. Since there is no second-order statistic for $v(n)$, we replace the traditional power with a generalized power:

$$\gamma = \left(\frac{P_{\mathrm{dn}}}{10^{\mathrm{GSNR}/10}}\right)^{\frac{1}{\alpha}}, \quad \beta = \delta = 0, \tag{32}$$

where P_{dn} denotes the power of $d(n)$ and GSNR is the generalized SNR. By adjusting the values of α and GSNR, noises with diverse pulse intensities can be generated. The experimental parameters are set with $\alpha \in (1.0, 2.0)$ in increments of 0.2, and GSNR $\in (5, 15)$ in increments of 5. The power of direct FECG $s(n)$ and the maternal ECG component $m(n)$ are fixed. Through varying the values of α and GSNR, the power of the noise $v(n)$ is adjusted, allowing for the regulation of the SNR between $s(n)$ and $v(n)$. For each combination of noise parameters, 100 sequences are generated as noise sources. These noises are stored for Monte Carlo simulations aimed at determining the optimal parameters for each algorithm.

D. Results on synthetic data

Prior to the extraction of FECG, we first establish the optimal matching parameters between subjects and algorithms through Monte Carlo simulations. The parameter ranges are defined as follows: filter order $L \in (0, 250)$ and filter step size $\mu \in (0, 1)$.

Table III presents the simulation results for several subjects under their respective optimal matching parameters. The results indicate that, regardless of the presence of Gaussian noise ($\alpha = 2.0$) or non-Gaussian noise ($\alpha < 2.0$), the performance of FECG extraction using Llncosh, LHSAF, and ILHSAF algorithms is significantly better than that of NLMS and IPNLMS algorithms. This is particularly evident in the presence of impulsive noise, suggesting that MSE-based algorithms struggle to maintain their performance under such conditions. The proposed LHSAF and ILHSAF algorithms generally demonstrate superior accuracy compared to Llncosh, as evidenced by lower RMSE and PRD values, in addition to higher SNR. Furthermore, the extraction accuracy of the ILHSAF algorithm is sometimes slightly inferior to that of the LHSAF algorithm, which may be attributed to a trade-off between fast convergence and stability.

Fig. 6 illustrates the overlap between FECG and MECG from the abdominal signal of the mother, identified as subject-01 in the FECGSYN dataset. Three types of overlap are considered, as shown in the figure, the red circle indicates a complete overlap between FECG and MECG, the green circle indicates no overlap, and the black circle denotes partial overlap.

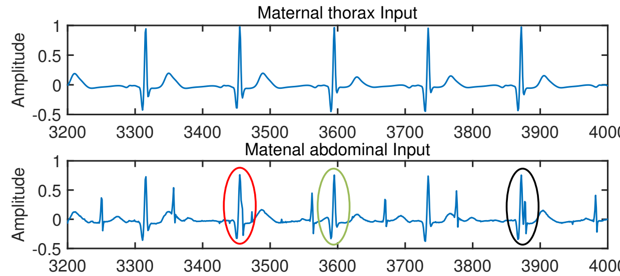


Fig. 6. MECG and AECG recordings of subject-01 from FECGSYN. The red circle denotes complete overlap, the green circles denotes no overlap, and the black circle denotes partial overlap.

Fig. 7 shows the graphical responses of the FECG extraction for subject-01, with $\alpha = 2.0$ and $\alpha = 1.4$. It can be observed that all algorithms successfully separate the FECG signal in both Gaussian and non-Gaussian noise backgrounds. However, their effectiveness in eliminating the maternal ECG component varies. As illustrated in Fig. 7(a), when $\alpha = 2.0$, both the IPNLMS and ILHSAF algorithms effectively remove the maternal ECG component $m(n)$, while the outputs of the Llncosh and LHSAF algorithms exhibit residual $m(n)$, as highlighted by the orange and green dashed circles. When α is reduced to 1.4, the noise becomes more impulsive with larger outlier values. In this scenario, the output of IPNLMS still retains residual $m(n)$, as indicated by the red dashed circle in Fig. 7(b), while the performance of Llncosh and LHSAF algorithms improve. Notably, the ILHSAF algorithm remains effective in eliminating the redundant components under both noise conditions.

E. Results on real data

Fig. 8 presents the AECG signals from channels 1, 2, 3, and 5 of a single subject in the Daisy dataset. Channel 4 is excluded from further analysis due to severe contamination by fetal peak information, which could compromise algorithm performance [52]. As shown in Fig. 8, red dashed circle indicates complete overlap, green dashed circle denotes no overlap, and black dashed circle represents partial overlap.

Since the Daisy dataset does not provide fetal peak annotations, we use manual annotations as a reference [52].

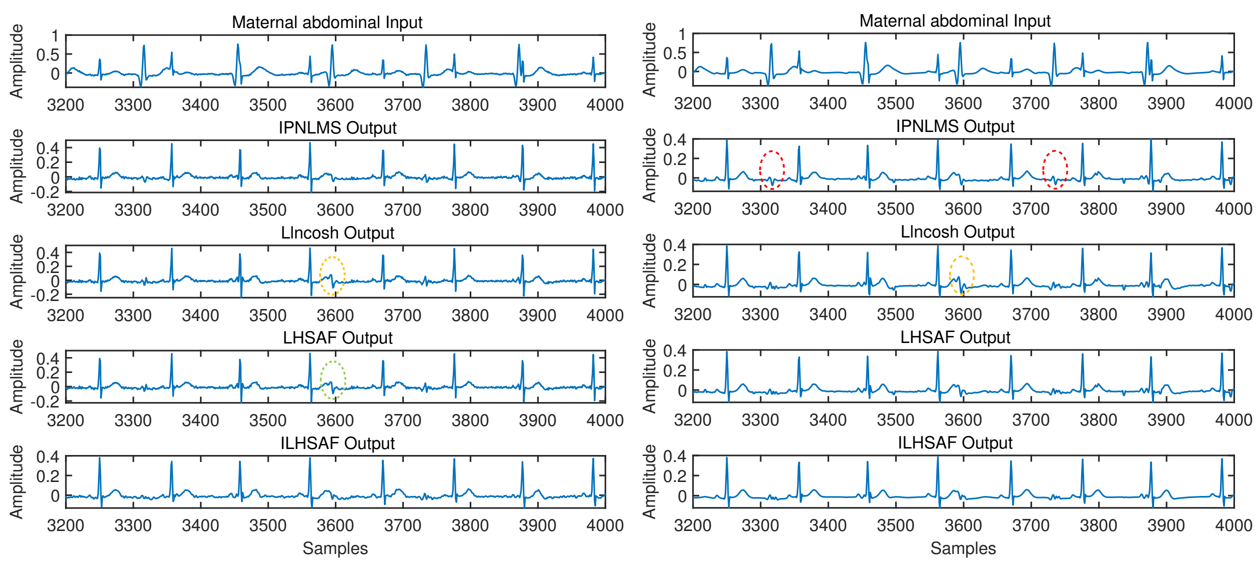
Table IV presents the statistical analysis of the proposed ILHSAF algorithm, where all metrics reach 100%. It means all peaks including the overlapped and partially overlapped signals are successfully detected, demonstrating the high accuracy and effectiveness of the algorithm.

Fig. 9 illustrates the graphical responses of several algorithms when $\alpha = 1.4$. As shown in Fig. 9, the NLMS algorithm fails to filter out maternal ECG components, as indicated by the red dashed circles. Although IPNLMS shows better performance than NLMS, it still encounters similar challenges, as highlighted by the orange dashed circles.

Both the Llncosh algorithm and the proposed ILHSAF algorithm effectively remove maternal ECG components under strong impulsive noise, exhibiting superior filtering performance. However, the Llncosh algorithm is constrained by a step-size limitation: when $\mu > 0.2$, the output waveform becomes distorted, resulting in unstable filtering performance.

TABLE III
OPTIMAL EVALUATION RESPONSE FOR EACH ALGORITHM WITH FECGSYN

Subject	RMSE / SNR(dB) / PRD(%) under GSNR = 15dB					
	α	NLMS	IPNLMS	Llncosh	LHSAF	ILHSAF
Sub01	2.0	4.338/4.542/64.430	3.195/6.444/49.520	2.880/7.238/46.053	**2.865/7.295/45.895**	2.886/7.297/45.901
	1.8	4.396/4.488/65.666	3.282/6.348/51.190	2.944/7.170/47.365	**2.924/7.201/47.149**	2.928/7.202/47.197
	1.6	4.489/4.436/68.857	3.368/6.306/54.317	3.055/7.106/50.690	**3.024/7.136/50.483**	3.029/7.138/50.527
	1.4	4.850/4.221/87.190	3.763/6.011/72.313	3.526/6.701/70.288	**3.504/6.734/70.058**	3.510/6.734/70.115
Sub03	2.0	2.290/2.458/95.124	1.405/3.732/77.591	1.086/4.317/72.093	**1.085/4.331/72.081**	1.085/4.328/72.083
	1.8	2.361/2.458/98.658	1.445/3.793/80.903	1.072/4.448/74.700	1.067/4.435/74.620	**1.067/4.442/74.615**
	1.6	2.399/2.519/105.396	1.423/3.901/87.232	1.089/4.581/82.238	**1.083/4.599/82.144**	1.083/4.609/82.144
	1.4	2.868/2.371/150.117	1.987/3.326/88.695	1.635/4.376/88.295	1.626/4.405/88.150	**1.625/4.409/88.129**
Sub05	2.0	2.489/4.313/70.844	1.723/5.492/59.389	1.424/6.006/55.439	1.424/6.006/55.440	**1.424/5.989/55.434**
	1.8	2.557/4.296/73.237	1.765/5.529/61.638	1.451/6.064/57.558	1.450/6.043/57.548	**1.450/6.061/57.543**
	1.6	2.640/4.304/78.830	1.816/5.599/66.991	1.509/6.103/63.120	1.507/6.112/63.101	**1.507/6.121/63.100**
	1.4	3.073/4.094/110.142	2.278/5.339/98.329	2.068/5.769/97.272	2.068/5.774/97.276	**2.068/5.783/97.274**
Sub07	2.0	2.920/4.290/71.944	1.736/6.097/54.688	0.958/7.563/45.728	**0.937/7.604/45.509**	0.938/7.624/45.515
	1.8	2.990/4.225/74.058	1.807/6.073/57.152	0.910/7741/47.176	**0.879/7.803/46.872**	0.880/7.827/46.880
	1.6	3.059/4.262/78.945	1.809/6.192/61.201	0.898/7.928/51.911	**0.854/8.016/51.512**	0.863/7.996/51.594
	1.4	3.489/4.047/108.519	2.263/5.930/90.425	1.428/7.618/83.291	**1.380/7.716/82.878**	1.405/7.691/83.083
Sub09	2.0	1.822/1.288/121.258	1.548/1.547/113.863	**0.834/2.996/96.590**	0.837/3.008/96.659	0.843/2.994/96.799
	1.8	1.884/1.308/126.333	1.608/1.625/119.055	**0.788/3.177/99.979**	0.793/3.171/100.057	0.798/3.168/100.162
	1.6	1.896/1.382/135.693	1.604/1.673/128.119	0.790/3.373/110.839	**0.788/3.369/110.785**	0.794/3.362/110.833
	1.4	1.389/1.312/199.865	1.106/1.603/192.345	1.303/3.342/178.866	**1.303/3.341/178.773**	1.312/3.329/178.914



(a) $\alpha = 2.0$ (b) $\alpha = 1.4$

Fig. 7. Graphical responses of FECG extraction with subject-01 under different noise background.

TABLE IV
RESULTS OF R-PEAK DETECTION PERFORMANCE ANALYSIS WITH THE DAISY DATASET.

Recording	Numbers	TP	FP	FN	Se(%)	PPV(%)	F1(%)
1	22	22	0	0	100	100	100

The signals in the NI-FECG dataset are weak, and the fetal ECGs are severely contaminated. To this end, we use 18 recordings from the NI-FECG and ensure that the abdominal channel signal selected for each recording allows for the observation of QRS complex. Each abdominal channel intercepts a 10s data segment. The analysis of Se, PPV and F1-score of the selected recordings is presented in Table V. Five recordings from the NI-FECG dataset have no FP or FN, with all three metrics reaching 100%. Performance fluctuations observed in the remaining recordings may be attributed to differences in the intrinsic quality of the signals [12]. The proposed ILHSAF algorithm demonstrates strong performance

TABLE V
RESULTS OF R-PEAK DETECTION PERFORMANCE ANALYSIS WITH THE NI-FECG DATASET.

Database	Recordings	TP	FP	FN	Number of Fetal R-peak in annotations	Se	PPV	F1
NI-FECG	ecgca102	14	0	1	15	93.3	100	96.6
	ecgca115	14	2	0	14	100	87.5	93.3
	ecgca252	15	0	1	16	93.8	100	96.8
	ecgca274	15	0	0	15	**100**	**100**	**100**
	ecgca323	15	1	0	15	100	93.8	98.8
	ecgca392	13	0	3	16	81.3	100	89.7
	ecgca416	14	1	1	15	93.3	93.3	93.3
	ecgca445	11	1	4	15	73.3	91.7	81.5
	ecgca515	16	0	0	16	**100**	**100**	**100**
	ecgca597	14	1	0	14	100	93.3	96.6
	ecgca621	14	1	1	15	93.3	93.3	93.3
	ecgca649	16	0	0	16	**100**	**100**	**100**
	ecgca733	12	2	1	13	92.3	85.7	88.9
	ecgca776	15	1	0	15	100	93.8	96.8
	ecgca864	16	0	0	16	**100**	**100**	**100**
	ecgca880	13	1	2	15	86.7	92.9	89.7
	ecgca902	15	0	0	15	**100**	**100**	**100**
	ecgca986	14	0	2	16	87.5	100	93.3
Total		256	11	16		-	-	-
Average		-	-	-		94.2	95.9	94.8

TABLE VI
COMPARISON OF OUTCOMES WITH OTHER EXISTING METHODS.

Algorithms	Database	Recording	Se	PPV	F1	Database	Recordings	Se	PPV	F1
NLMS [19]	Daisy	1	68.4	81.3	74.3	NI-FECG	ecgca115 ecgca274 ecgca323 ecgca445 ecgca515 ecgca649 ecgca776 ecgca880 ecgca902	92.3	95.3	93.6
IPNLMS [20]			85.1	60	70.1			81.2	95.3	86
Llncosh [21]			100	95.7	97.8			87.3	93.8	89.5
Proposed method			100	100	100			95.6	95.5	95.3

with all metrics exceeding 80% for all recordings except for ecgca445. Moreover, the FECG signals extracted for R-peak detection yield an average Se of 94.2%, an average PPV of 95.9%, and an average F1-score of 94.8%.

Fig. 10 displays the graphical responses of the proposed ILHSAF algorithm compared to several other algorithms with the ecgca274 recording under strong impulsive noise conditions($\alpha = 1.4$). The results indicate that the proposed method is capable of extracting FECG information to the greatest extent compared to other algorithms, showcasing its effective filtering performance and good robustness.

Table VI compares the proposed ILHSAF method with other AF algorithms using the same datasets. For the Daisy dataset, the NLMS algorithm yields the lowest Se value, achieving only 68.4%. IPNLMS exhibits the lowest PPV and F1-score, with values of 60% and 74.3%, respectively. The performance of Llncosh ranks second to that of our proposed method, achieving a Se value of 100%. For the NI-FECG dataset, we conduct a performance analysis on the same recordings, and the statistical metrics shown in Table VI are their average values. It can be concluded that the performance of NLMS, IPNLMS, and Llncosh algorithms is unstable, making it difficult to distinguish which one is superior. Only the proposed ILHASF algorithm consistently performs the best on the used dataset, suggesting that it possesses both high stability and efficacy in FECG extraction.

F. Analysis of PQRST waves

A typical cycle of the FECG signal comprises a series of characteristic waves: P wave, Q wave, R wave, S wave, and T wave, each of them has different physiological significance. The complete PQRST complex provides comprehensive insights into fetal cardiac health, facilitating timely clinical diagnosis and intervention [53]. Since the FECGSYN offers the original FECG signals, we evaluate the quality of the reconstructed FECG by observing the degree of overlap between the original and the extracted FECG waveforms. Fig. 11 presents the visual error in waveform response for subject-01 when $\alpha = 1.4$, with the original signal indicated in red. Upon visual inspection, it is evident that the extracted FECG closely resembles the original FECG signal. We choose samples from 3300 to 3600, and based on the annotations, the positions of the PQRST waves are shown in Fig. 11. The

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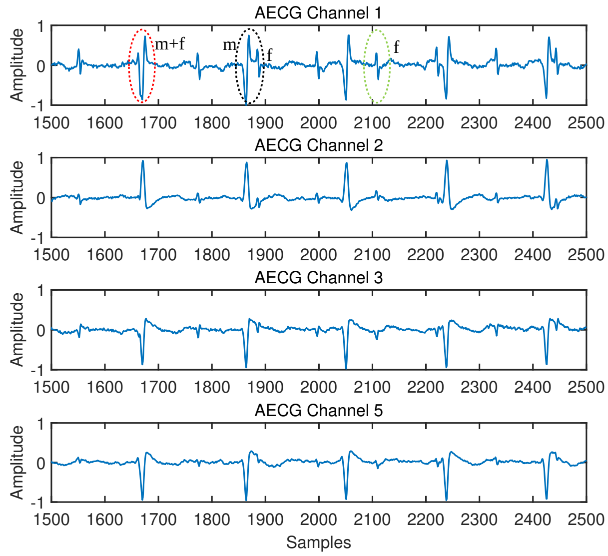

Fig. 8. AECG recordings from 4 channels of a subject in the Daisy dataset. The red dashed circle indicates complete overlap, the black dashed circle represents partial overlap, and the green dashed circle denotes no overlap.

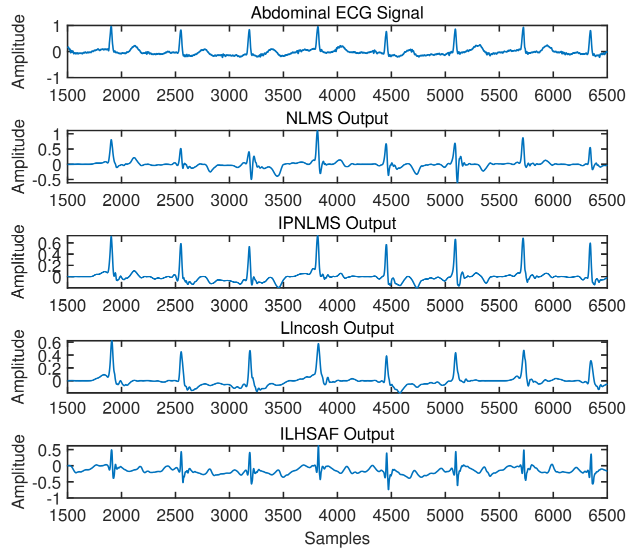

Fig. 10. Graphical responses of FECG extraction for recording ecgca274 under impulsive noise with $\alpha = 1.4$.

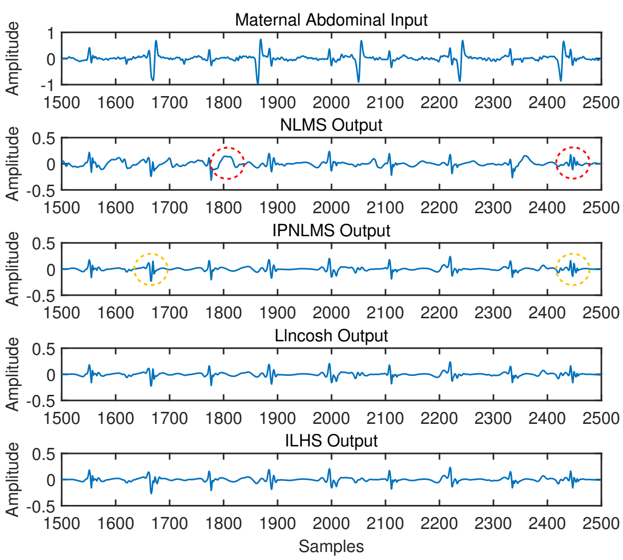

Fig. 9. Graphical responses of FECG extraction for Daisy dataset under impulsive noise with $\alpha = 1.4$.

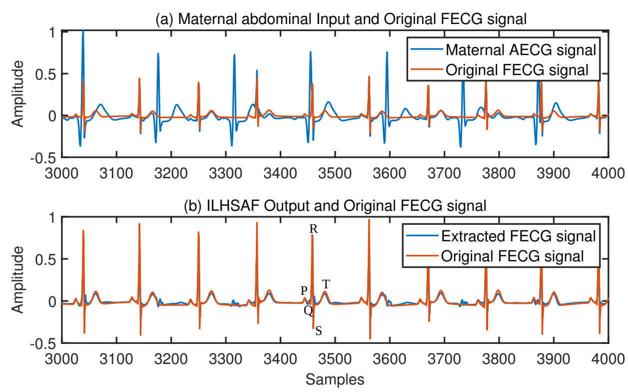

Fig. 11. Visual error evaluation of original and extracted FECG signals.

full-wave positions of the extracted FECG closely align with those of the original FECG, indicating that the FECG extracted by ILHSAF algorithm exhibits good waveform integrity. Due to the absence of original FECG signals and corresponding annotations in the real-world databsets Daisy and NI-FECG, we use manual annotations as a reference. Fig. 12 and Fig. 13 respectively show the full-wave positions from Daisy and ecgca274 in NI-FECG. In both figures, the approximate locations of PQRST waves can be clearly identified, further validating the feasibility of the proposed algorithm for non-invasive FECG extraction.

V. Conclusions

This study proposes the LHSAF algorithm based on a novel natural logarithmic hyperbolic secant function for NIFECG extraction. By introducing a hyperbolic tangent-like transformation, the ILHSAF algorithm is derived for parameter optimization and FECG information preservation. The proposed methods are evaluated in three ways under α-stable distribution noise. First, the graphical response of the FECG extracted from the AECG is assessed. Second, the fetal QRS detection of the extracted FECG is evaluated. Finally, the full-wave analysis of the extracetd is conducted. We present FECG extraction results on one synthetic dataset and two real-world datasets. The proposed ILHSAF algorithm consistently outperforms the mentioned AF methods, demonstrating significant improvements in the aspects of accuracy, convergence and robustness. Specifically, in comparison with NLMS, IPNLMS and Llncosh algorithms, the ILHSAF algorithm achieves 100% in all three metrics on the Daisy dataset. On the NI-FECG dataset, it attains Se, PPV, and F1-score of 95.6%, 95.5%, and 95.3%, respectively. As a result, the proposed method can be regarded as a valuable tool for continuous fetal heart monitoring, as the method operates in real-time.

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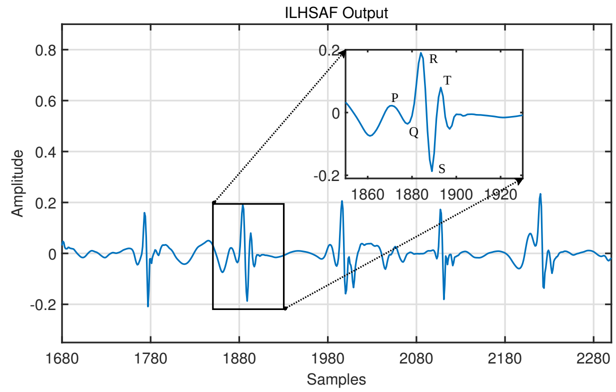

Fig. 12. PQRST waves of the graphical response from Daisy through ILHSAF.

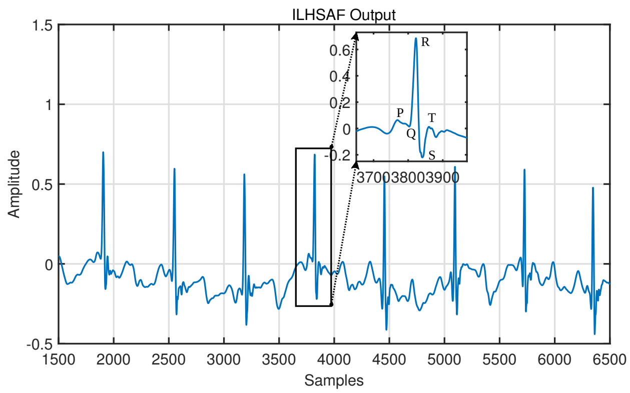

Fig. 13. PQRST waves of the graphical response of ecgca274 in NI-FECG obtained by ILHSAF.

Appendix
Proof of Step Size Conditions for Stability Analysis

The stability and convergence of AF algorithms are dependent on the selection of the step-size parameter μ [24]. For a robust adaptive controller, the optimal weight vector is denoted as $\boldsymbol{w}_o$, the deviation between the weight vector $\boldsymbol{w}(n)$ at time n and $\boldsymbol{w}_o$ is given by:

$$\tilde{\boldsymbol{w}}(n) = \boldsymbol{w}_o - \boldsymbol{w}(n), \tag{1}$$

the weight update rule of LHSAF is given as:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu \frac{\tanh(\lambda e(n))\,\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \boldsymbol{x}(n). \tag{2}$$

Subtract $\boldsymbol{w}_o$ from both sides of (2) yields:

$$\begin{aligned} \boldsymbol{w}(n+1) - \boldsymbol{w}_o = \boldsymbol{w}(n) - \boldsymbol{w}_o + \\ \mu \frac{\tanh(\lambda e(n))\,\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \boldsymbol{x}(n), \end{aligned} \tag{3}$$

thus, for LHSAF, the recurrence formula for the deviation between $\tilde{\boldsymbol{w}}(n)$ and $\boldsymbol{w}_o$ can be written as:

$$\tilde{\boldsymbol{w}}(n+1) = \tilde{\boldsymbol{w}}(n) - \mu \frac{\tanh(\lambda e(n))\,\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \boldsymbol{x}(n). \tag{4}$$

Taking expectation of the squared Euclidean norm on both sides of (4), we get

$$\begin{aligned} D(n+1) = D(n) + \mu^2 \mathrm{E}\left\{ (\tau\left[e(n)\right])^2 \|\boldsymbol{x}(n)\|^2 \right\} \\ - 2\mu \mathrm{E}\left\{ \tau\left[e(n)\right] \tilde{\boldsymbol{w}}^{\mathsf{T}}(n) \boldsymbol{x}(n) \right\}, \end{aligned} \tag{5}$$

where, $\tau[e(n)] = \frac{\tanh(\lambda e(n))\,\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))}$, and $D(n) = \mathrm{E}\left[\|\tilde{\boldsymbol{w}}(n)\|^2\right]$ denotes the expectation of the squared Euclidean norm for deviation of the weight vector. To achieve stability and convergence of the algorithm, the condition that $D(n+1)$ is smaller than $D(n)$ should be satisfied. Hence, the range of proper step size μ_1 can be derived as follows:

$$\begin{aligned} & D(n+1) - D(n) < 0 \\ \Rightarrow\ & {\mu_1}^2 \mathrm{E}\left\{ (\tau\left[e(n)\right])^2 \|\boldsymbol{x}(n)\|^2 \right\} < 2\mu_1 \mathrm{E}\left\{ \tau\left[e(n)\right] \tilde{\boldsymbol{w}}^{\mathsf{T}}(n) \boldsymbol{x}(n) \right\} \\ \Rightarrow\ & {\mu_1}^2 \mathrm{E}\left\{ \left(\frac{\tanh(\lambda e(n))\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \right)^2 \|\boldsymbol{x}(n)\|^2 \right\} \\ & < 2\mu_1 \mathrm{E}\left\{ \frac{\tanh(\lambda e(n))\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \tilde{\boldsymbol{w}}^{\mathsf{T}}(n) \boldsymbol{x}(n) \right\} \\ \Rightarrow\ & 0 < \mu_1 < \frac{2\mathrm{E}\left\{ \frac{\tanh(\lambda e(n))\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} |e(n)| \right\}}{\mathrm{E}\left\{ \left(\frac{\tanh(\lambda e(n))\mathrm{sech}(\lambda e(n))}{1+\mathrm{sech}(\lambda e(n))} \right)^2 \|\boldsymbol{x}(n)\|^2 \right\}} \end{aligned} \tag{6}$$

where we assume that $\tilde{\boldsymbol{w}}(n)^{\mathsf{T}} \boldsymbol{x}(n) = e(n)$.

The weight update rule of ILHSAF is expressed as:

$$\begin{aligned} \boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \\ \mu \frac{\tanh(\frac{0.35}{\mathrm{H}} e(n))\,\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))}{1+\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))} \boldsymbol{x}(n), \end{aligned} \tag{7}$$

similarly, we can derive that the proper step-size range of ILHSAF is:

$$0 < \mu_2 < \frac{2\mathrm{E}\left\{ \frac{\tanh(\frac{0.35}{\mathrm{H}} e(n))\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))}{1+\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))} |e(n)| \right\}}{\mathrm{E}\left\{ \left(\frac{\tanh(\frac{0.35}{\mathrm{H}} e(n))\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))}{1+\mathrm{sech}(\frac{0.35}{\mathrm{H}} e(n))} \right)^2 \|\boldsymbol{x}(n)\|^2 \right\}} \tag{8}$$

Furthermore, analysis of the weight update rules (2) and (7) reveals that their core nonlinear terms incorporate $\tanh$ and sech functions. The saturation behavior of the $\tanh(\cdot)$ and the rapid decay characteristic of the $\mathrm{sech}(\cdot)$ inherently constrain the magnitude of the weight updates. This mechanism effectively suppresses the disruptive impact of large errors induced by impulsive noise, contributing significantly to the algorithm's stability.

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